

Gulbarga Anirudh Rao

Cloud Infra Engineer | Network Virtualization | AWS Certified SAA

940-300-2958 | AnirudhRaoGulbarga@my.unt.edu | linkedin.com/in/anirudh | github.com/1-Anirudh

SUMMARY

Computer Science graduate student and AWS Certified Solutions Architect – Associate, with hands-on experience in hybrid cloud infrastructure, network programming, and protocol debugging. Skilled in designing and deploying scalable solutions using core AWS services like EC2, S3, IAM, VPC, RDS, CloudWatch, and Lambda. At LTIMindtree, improved performance across 200+ Cisco routers using Python automation with Zabbix, Splunk, and Wireshark. Proficient in BGP, OSPF, EIGRP, system administration, and multi-cloud platforms (AWS, GCP). Strong foundation in distributed systems, WebSockets, and full-stack development. Seeking roles in cloud networking, infrastructure automation, or technical support.

EDUCATION

University of North Texas

Master of Science in Computer Science

Denton, TX

Aug. 2024 – Present

Mahatma Gandhi Institute of Technology

Bachelor of Technology in Electrical Engineering

Hyderabad, India

Aug. 2018 – May 2022

EXPERIENCE

Cloud & Infrastructure Engineer

LTIMindTree

Sept 2023 – Jul 2024

Pune, India

- Designed hybrid-cloud monitoring using **Zabbix & Splunk**; automated workflows with Python, improving uptime by **30%**
- Built Python diagnostic tool to parse **BGP/OSPF** logs across 200+ Cisco routers, cutting triage time by **40%**
- Resolved route flaps (**BGP/OSPF/EIGRP**) in HA networks, ensuring sub-second failover and zero customer disruption
- Managed 50+ servers (**Linux/Windows**) with patching, access control, and uptime SLA enforcement
- Analyzed **1M+** packets via **Wireshark & Splunk** to trace OSI-layer issues and reduce packet loss by **25%**
- Automated alerts using Python scripts with Zabbix & **ServiceNow** APIs, reducing MTTR by **30%**
- Collaborated to design and deploy secure hybrid cloud solutions, ensuring seamless data flow between on-premise and cloud

Networking Intern

LTIMindTree

Jun 2023 – Aug 2023

Pune, India

- Configured and troubleshoot **Cisco routers/switches**; gained hands-on experience with OSPF, BGP, EIGRP, and MPLS
- Applied core networking protocols (**TCP/IP, SIP, SCCP, MGCP**) in VoIP and secure communication labs
- Supported **Windows/Linux** system admin tasks, including patching, service monitoring, and user account provisioning
- Conducted network diagnostics using Cisco CLI and **SolarWinds**, contributing to early-stage performance and analysis

KEY PROJECTS

Secure Network Protocol Analysis using Wireshark | *Wireshark, TCP/IP, TLS, DNS, HTTP, ICMP*

Fall 2024

- Captured and thoroughly analyzed multi-protocol network traffic (ICMP, TCP, UDP, DNS, HTTP, HTTPS) using Wireshark in a Linux environment for deep packet inspection and comprehensive network troubleshooting across multiple OSI layers
- Enhanced packet inspection via advanced filters to analyze TCP handshakes, TLS encryption, and HTTP streams
- Generated synthetic test traffic using tools like **ping**, **traceroute**, and **nslookup** to validate protocol behaviors and latency
- Investigated DNS A/MX/TXT queries and thoroughly evaluated UDP vs. TCP fallback for resilient name resolution
- Assessed end-to-end latency and packet loss across **traceroute** hops to identify bottlenecks and router-level timeouts

Research Finder | *Node.js, Express.js, EJS, Firebase, WebRTC, JavaScript, HTML/CSS, Vercel*

Fall 2024

- Built and deployed a scalable research platform with **EJS**, **Node.js**, and **Firebase**, compatible across modern browsers
- Enhanced accessibility through **voice navigation**, **screen reader** support, and a **multilingual interface** for user groups
- Developed a recommendation system using **Pinecone** to suggest roles based on users' skills, interests, and activity history
- Enabled collaboration via **secure messaging**, **video calls**, and real-time alerts, **Authentication**, feedback and posts
- Authored documentation, used **Firebase Analytics** to track engagement, and deployed on **Vercel** with admin dashboards

Fast Chat | *Python, PostgreSQL, WebSockets, Socket Programming, E2E RSA Encryption*

Fall 2024

- Built **E2E Encrypted** chat app for secure, authenticated communication using **Python Sockets** and **PostgreSQL**
- Implemented features including **private and group chats**, **admin roles**, and support for **text and image sharing**
- Enhanced performance by creating a **load server** for load distribution and a **master server** for server communication
- Implemented **RSA**-based key exchange, generating per-session keys and storing encrypted session tokens in **PostgreSQL**

StockVision | *Python, TensorFlow, Keras, Scikit-learn, NLTK, Transformers, Matplotlib, MinMaxScaler* Fall 2024

- Developed an **LSTM-based** model for Tesla stock price prediction, integrating **momentum indicators**, **moving averages**, volatility-based risk, news sentiment features, and cross-validation testing to capture complex temporal patterns
- Used **NLP transformers** for financial news sentiment analysis, improving prediction accuracy and true positive rate
- Optimized preprocessing with **feature engineering**, **MinMaxScaler**, and **dropout layers** to prevent overfitting
- Compared model performance to **XGBoost**, showcasing **LSTM's** superior **sequence learning** and **scalability**

GlucoSight | *Python, scikit-learn, pandas, NumPy, Matplotlib, Flask, HTML/CSS* Spring 2025

- Developed an end-to-end ML pipeline for the Pima Indians Diabetes dataset using **statistical imputation**, **MinMax scaling**, incorporating **feature engineering**, **hyperparameter tuning** to ensure data integrity and production readiness
- Performed feature selection flagging Glucose, BMI, Age as key predictors; **reducing dimensionality** with **98.7%** accuracy
- Benchmark-tested six classifiers, hyperparameter-tuned **KNN** and **RF**, and deployed an **SVM** for best-in-class performance
- Established rigorous evaluation framework using train-test splits, precision-recall, and **confusion matrix** for reliability
- Deployed the model with a **Flask** web app and **/predict REST API**, supporting **real-time** diabetes risk predictions

Cryptonite | *Python, SimPy, NetworkX, Matplotlib, Git, GitHub* Fall 2024

- Developed a **discrete time event simulator** modelling a **blockchain cryptocurrency network** with **honest miners**
- Simulated **message latency**, **proof of work**, **transaction generation**, **verification** and **blockchain forks**
- Visualized **blockchain states** and **mining activities**, delivering **real-time** analysis and insights into **network dynamics**
- Implemented dynamic network topology, simulating peer connections and testing **scalability** under varied node parameters

OTHER PROJECTS

Learning Engine | *Python, SQLite, SQL* Fall 2024

- Built Python SQLite university data app where student can view/add/enroll features, streamlined admin tasks.
- Designed an interface with a main menu and clear sub-menus, enhancing navigation across database and user experience
- Implemented data validation, error handling, and transaction techniques to ensure integrity and secure operations
- Authored detailed **documentation** outlining setup, configuration, and execution steps, enhancing usability and support

Autonomous Driving Model | *Python, TensorFlow, Keras, CNN, OpenCV, NVIDIA DAVE-2* Spring 2025

- Implemented **OpenCV** preprocessing (**cropping**, **normalization**) to reduce noise and improve **CNN** training
- Trained a neural network model for autonomous driving using advanced **CNNs** and **behavioral cloning techniques**
- Captured and augmented extensive data from **Udacity's simulator** using **three cameras**, improving model accuracy
- Modified and optimized **NVIDIA's DAVE-2 architecture** to significantly enhance lane-following performance

Deep Face Recognition Model | *Python, TensorFlow, Keras, NumPy, OpenCV, SGD Optimizer* Spring 2025

- Surveyed literature on state-of-art models like **DeepFace**, **FaceNet**, **ArcFace** to learn discriminative feature vectors
- Implemented the **FaceNet** paper in **Python** and **TensorFlow** to extract highly discriminative feature vectors
- Trained the **FaceNet** model using **SGD** on the **LFW dataset**, which includes **13.4K+ images** of **5K+ people**

CERTIFICATIONS

- **AWS Certified Solutions Architect** – Associate, Amazon Web Services (Credential ID: SAA-C03, active) [Certificate]
- **AWS Cloud Support Associate Professional Certificate**, Coursera (Issued by Amazon Web Services) [Certificate]

7-course specialization: IT fundamentals, core AWS services, troubleshooting, automation, serverless, and customer support

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript, HTML, Ejs, CSS, C++, C, Bash, $\text{\LaTeX}$

Cloud & Devops: GCP (VPC, Monitoring, Compute), Docker, Firebase, Vercel, Amazon(VPC, EC2, S3, RDS, CloudWatch, Route 53), AWS(IAM, Lambda, Auto Scaling, CloudFront, CloudFormation, CLI, SDKs), Elastic Load Balancing (ELB)

Networking & Monitoring: Wireshark, Zabbix, BGP/OSPF/EIGRP, Cisco CLI, Splunk, ServiceNow, SolarWinds

Developer Tools: Git, Docker, TravisCI, Google Cloud Platform, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse

Frameworks: Node.js, Express.js, Flask, React, FastAPI, EJS, WordPress, Material-UI

Libraries: scikit-learn, pandas, NumPy, Matplotlib, Pinecone, TensorFlow, OpenCV, WebRTC, Socket Programming

RELEVANT COURSEWORK

Computer Science Data Structures & Algorithms, Big Data, Data Science, Fundamentals of Database, Software Engineering, Computer Algorithms, Software development for AI, Computer Networks, Operating Systems Design*, AWS Cloud Support Associate

**Ongoing*

EXTRACURRICULAR ACHIEVEMENTS

- Participated in Poster Presentation on Environmental Biotechnology held at AMS, Osmania University with distinction
- Participated in one day national level seminar on Basic Sciences – Drug Development and Applications research insights
- Attended Computer Basic Training conducted by Member of Parliament Local Area Development program initiative
- Participated in one day workshop on personality development organized by National Service Scheme youth program
- Participated in summer special camp held by National Service Scheme focusing on rural community development